

LE ANH DUY

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EDUCATION

VNUHCM - University of Science (HCMUS), Ho Chi Minh City, Vietnam Enrolled: 2023 — Expected: 2027
Bachelor of Science in Information Technology GPA: 9.08 / 10.0 — 3.83 / 4.00
Core CS Coursework (Perfect 10.0): DSA, Data Science, Linear Algebra, Discrete Mathematics, Intro to Computer Vision.
Advanced AI & Foundations: Machine Learning (9.6), Fundamentals of AI (9.3), Object-Oriented Programming (9.4).

ACHIEVEMENTS

- **Silver Award** – ICPC Southern Area National Programming Contest 2024 (Team: HCMUS-TrioProject).
- **Honorable Prize** – Vietnam Olympiad in Informatics (VOI) 2022 & 2023.
- **First Runner-Up** – Thach Thuc 2026 Academic Competition, HCMUS.
- **81st Place** – ICPC National Programming Contest 2026.
- **Honorable Prize** – ICPC Southern Area National Programming Contest 2026.

PUBLICATIONS & RESEARCH PAPERS

Egis: AI-Enhanced OSINT for Multimedia Verification **Paper Link – 2025**

Co-author — *Published in Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*

- Developed a semi-automated multimedia verification paradigm submitted to the ACM MM'25 Grand Challenge
- Engineered the hybrid geolocation module utilizing deterministic heuristic and rule-based baseline methods to establish a robust non-learning benchmark for cross-platform geospatial data alignment.
- Evaluated localization accuracy across complex conflict-zone multimedia datasets by cross-referencing spatial artifacts with Google Earth and Street View pipelines.

MERVIN: A Unified Framework for Multimodal Event Retrieval in Vietnamese News Videos **Paper Link – 2026**

Co-author — *Accepted at the Symposium on Information and Communication Technology (SOICT)*

- Co-designed a unified cross-modal video search engine integrating keyframe visual representations and Vietnamese speech embeddings for semantic event localization.
- Formulated an LLM-based preprocessing workflow using Gemini 1.5 Flash to systematically filter acoustic distractions, dialectical variations, and errors in Automatic Speech Recognition (ASR) transcripts.
- Managed feature indexing inside a Milvus vector database, optimizing multimodal similarity-based text retrieval.

RESEARCH & AI PROJECTS

Human Action Recognition System using ST-GCN – Role: AI/Research Engineer **Project Link – 2026**

- Investigated deep spatial-temporal graph convolutional architectures for skeleton-based behavioral modeling on the FineGym dataset.
- Achieved a top validation accuracy of 89.68% and Macro F1 score of 87.19% through skeletal representation learning.
- Exposed a critical evaluation artifact in existing baseline literatures regarding truncated or incomplete sequence inputs.
- Counteracted extreme 8x structural class imbalances by integrating bone-vector geometry and deploying custom Focal Loss functions.
- Conducted ablation studies on network depth constraints, determining the optimal 6-block configuration for high-fidelity extraction.

Face Recognition Optimization via One-Shot NAS – Role: AI/Research Engineer **Project Link – 2026**

- Explored Neural Architecture Search (NAS) methodologies to automatically identify highly efficient, hardware-constrained face recognition topologies.
- Accelerated architectural search spaces from days to 1–2 hours by implementing a Single-Path One-Shot SuperNet backbone strategy.
- Streamlined exploration-phase compute costs by mapping architectural convergence directly from raw logits, bypassing expensive Average Precision (AP) iterations.
- Benchmarked and evaluated over 500 candidate models, verifying that optimization outcomes under constrained hardware closely match benchmarks established on high-end configurations.

FoodRecLSNet – End-to-End Visual Recommendation – Role: AI Engineer **Project Link – 2025**

- Managed multi-source dataset curation, structural cleaning, and exploratory data analysis for fine-grained localized image classification.
- Decoupled platform-dependent, low-level GPU tensor structures (CUDA/Triton) into pure hardware-agnostic NumPy mathematical routines to support portable evaluation.
- Evaluated generalization bounds across diverse convolutional backbones including ResNet, MobileNet, and EfficientNet variations.

TECHNICAL SKILLS

- **AI, Deep Learning & Vision:** PyTorch, Computer Vision (YOLO), Graph Neural Networks (ST-GCN), Neural Architecture Search (NAS), Multimodal Representation Learning, Model Evaluation & Ablation.
- **Data & Vector Infrastructure:** Milvus, FAISS, Data Preprocessing, Feature Engineering, Handling Severely Imbalanced Datasets.
- **Languages, Frameworks & Tools:** Python, C/C++, SQL, Git, Docker, FastAPI.
- **Core Competencies:** Competitive Programming (Trained Algorithmic Problem Solving), Open Source Intelligence (OSINT) Verification, Inference Optimization.